

Exercise Sheet 4

Principal component analysis

Use of AI You may use AI tools to help draft code, but your final answers should show your own judgment and explain why the results look like that.

Setup in Python Use `numpy`, `pandas`, and `matplotlib`. Use divisor m , the number of dates, throughout (`ddof=0`). For the SVD, use

```
np.linalg.svd(X, full_matrices=False).
```

For a symmetric covariance or correlation matrix, you can use `np.linalg.eigh`; it returns eigenvalues in increasing order.

Notation As in Lectures 3–4 and Sheet 3, put financial series in rows and dates in columns. For n series and m dates, let $\mathbf{R} \in \mathbb{R}^{n \times m}$, $\bar{\mathbf{r}} = \mathbf{R}\mathbf{1}_m/m$, and $\mathbf{X} = \mathbf{R} - \bar{\mathbf{r}}\mathbf{1}_m^\top$. For its rank- r SVD,

$$\mathbf{X} = \mathbf{U}_r \boldsymbol{\Sigma}_r \mathbf{V}_r^\top, \quad \hat{\mathbf{C}} = \frac{1}{m} \mathbf{X} \mathbf{X}^\top, \quad \mathbf{Y} = \mathbf{U}_r^\top \mathbf{X} = \boldsymbol{\Sigma}_r \mathbf{V}_r^\top.$$

The principal direction $\mathbf{u}_k \in \mathbb{R}^n$ gives the series loadings. Row k of $\mathbf{Y}$, written as a column, is the dated score series $\mathbf{y}_k = \mathbf{X}^\top \mathbf{u}_k = \sigma_k \mathbf{v}_k$; its variance is $\lambda_k = \sigma_k^2/m$.

Problem 1 Return to $\mathbf{A}$ from Sheet 3, Problem 1(a). Interpret its columns as two dates of returns for two assets, in percentage points:

$$\mathbf{R} = \mathbf{A} = \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}.$$

- Find $\bar{\mathbf{r}}$ and the centered matrix $\mathbf{X}$. Check that each row of $\mathbf{X}$ sums to zero.
- Compute $\hat{\mathbf{C}} = \mathbf{X} \mathbf{X}^\top / 2$. Find its eigenvalues and corresponding orthonormal eigenvectors by hand. Choose $\mathbf{u}_1$ to have positive entries.
- Use $\sigma_1 = \sqrt{2\lambda_1}$ and $\mathbf{v}_1 = \mathbf{X}^\top \mathbf{u}_1 / \sigma_1$ to obtain $\mathbf{X} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$. Compared with the uncentered SVD in Sheet 3, which of $\mathbf{u}_1$, σ_1 , and $\mathbf{v}_1$ have changed?
- Calculate the two scores in $\mathbf{y}_1 = \mathbf{X}^\top \mathbf{u}_1$. Interpret their signs and verify directly that their mean is zero and their variance is λ_1 . What fraction of total variance does PC1 explain?
- Reconstruct both original return columns using $\mathbf{r}_t = \bar{\mathbf{r}} + y_{1t} \mathbf{u}_1$.

Problem 2 Continue with the same financial data as Sheet 3, Problem 3. Reuse your uncentered log-return change matrix $\mathbf{R}$ and its dates, with rows `SPY`, `QQQ`, `TLT`, `HYG`, `VIX`, in that order. Here $n = 5$ and $m = T$, where T was the number of dates in Sheet 3.

- a) Verify that the shape, first and last dates, and row order of $\mathbf{R}$ agree with Sheet 3. Create the centered matrix $\mathbf{X}$ and the standardized matrix

$$\mathbf{Z} = \mathbf{D}^{-1}\mathbf{X}, \quad \mathbf{D} = \text{diag}(s_1, \dots, s_5), \quad s_i^2 = \frac{1}{m} \sum_{t=1}^m x_{it}^2.$$

Verify numerically that the rows of $\mathbf{Z}$ have mean zero and variance one.

- b) What role does centring play before PCA, and what additional modeling choice is made, when $\mathbf{X}$ is standardized to $\mathbf{Z}$?

Problem 3 Use the standardized data matrix $\mathbf{Z}$ from Problem 2.

- a) Compute $\mathbf{Z} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and the scores $\mathbf{Y} = \mathbf{U}^\top\mathbf{Z}$. Also compute $\hat{\mathbf{C}}_Z = \mathbf{Z}\mathbf{Z}^\top/m$ and its eigenpairs, ordered so that $\lambda_1 \geq \dots \geq \lambda_5$. Orient PC1 so that its score has a positive same-date correlation with SPY; orient each other component so that its largest absolute loading is positive. When changing the sign of an SVD singular vector, change both corresponding singular vectors. Align the covariance eigenvector signs with the SVD directions and verify

$$\lambda_k = \frac{\sigma_k^2}{m}, \quad \mathbf{u}_k^{(\text{cov})} = \mathbf{u}_k^{(\text{svd})}, \quad \mathbf{Z}^\top \mathbf{u}_k = \sigma_k \mathbf{v}_k.$$

- b) For equally shaped arrays $\mathbf{A}$ and $\mathbf{B}$, define their maximum absolute discrepancy by $d(\mathbf{A}, \mathbf{B}) = \max_{i,j} |A_{ij} - B_{ij}|$; for vectors, take the maximum over their entries. Compute this discrepancy between the two eigenvalue vectors, the two loading matrices, and the two score matrices in part a). Explain and verify the following equality: $\frac{1}{m}\mathbf{Z}\mathbf{Z}^\top = \mathbf{U}\frac{\mathbf{\Sigma}^2}{m}\mathbf{U}^\top$.
- c) Compute the following three quantities separately and verify that they agree:

$$\sum_{i=1}^5 \widehat{\text{Var}}(Z_i) = \frac{1}{m} \|\mathbf{Z}\|_F^2 = \sum_{k=1}^5 \lambda_k.$$

Define the individual and cumulative explained-variance fractions by

$$\text{EVR}_k = \frac{\lambda_k}{\sum_{j=1}^5 \lambda_j}, \quad \text{CEV}_K = \sum_{k=1}^K \text{EVR}_k.$$

Plot both. Choose the smallest K with $\text{CEV}_K \geq 80\%$, and report the fractions of total variance preserved and omitted.

- d) Plot the PC1 and PC2 loadings, score series from above, and interpret them.
- e) For PC1 and PC2, identify the date with the largest absolute score and explain its sign from the corresponding column of $\mathbf{Z}$.
- f) Use the same $\mathbf{R} \in \mathbb{R}^{5 \times m}$, dates, and row order in all three analyses:

matrix	analysis
$\mathbf{R}$	raw SVD from Sheet 3
$\mathbf{X} = \mathbf{R} - \bar{\mathbf{r}}\mathbf{1}_m^\top$	covariance PCA
$\mathbf{Z} = \mathbf{D}^{-1}\mathbf{X}$	correlation PCA.

For this comparison, orient each leading direction so that its dated coefficient series has positive same-date correlation with SPY. In a table: report the five leading directions coefficients, the first-component squared-energy fraction, and the smallest K whose cumulative fraction is at least 80%. Plot the three leading directions side by side and the three cumulative-fraction curves together. If there is a difference between the Sheet 3 and Sheet 4 results, are they caused by “SVD versus PCA,” or by the preprocessing choice?